1. Introduction to SQL

SQL (Structured Query Language) is a **programming language** used for **managing and manipulating data in relational databases**.
It allows performing **CRUD operations** (Create, Read, Update, Delete) in a database.
SQL is widely used in **applications, websites, businesses**, and also to **communicate with and control databases**.

2. Types of SQL

2.1 DDL (Data Definition Language)

Used to **create and modify databases or tables**.

Commands:

- `.create` – to create databases or tables
- `.alter` – to modify structure
- `.drop` – to remove databases or tables
- `.truncate` – removes all rows from a table

2.2 DML (Data Manipulation Language)

Used to **manipulate data in tables**.

Commands:

- `.create` – to insert new records (usually `INSERT`)
- `.delete` – to remove records
- `.update` – to modify records
- `.select` – to retrieve records

2.3 DCL (Data Control Language)

Used to **control access to data**.

Commands:

- `.grant` – provide permissions
- `.revoke` – remove permissions

2.4 TCL (Transaction Control Language)

Used to **manage transactions in databases**.

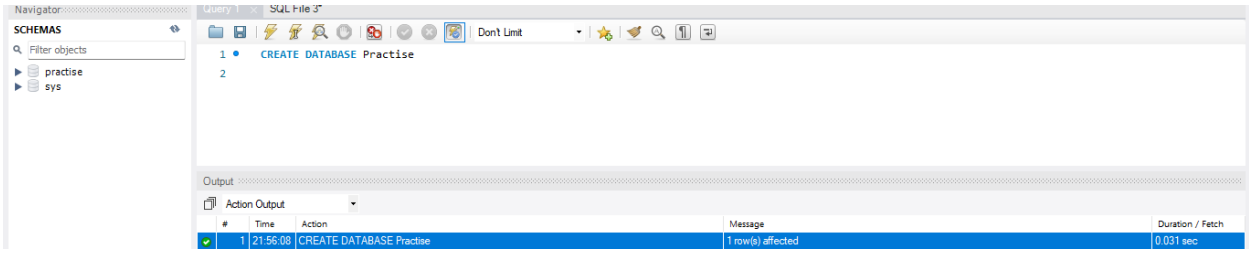
Commands:

- `.commit` – save changes permanently
- `.rollback` – undo changes

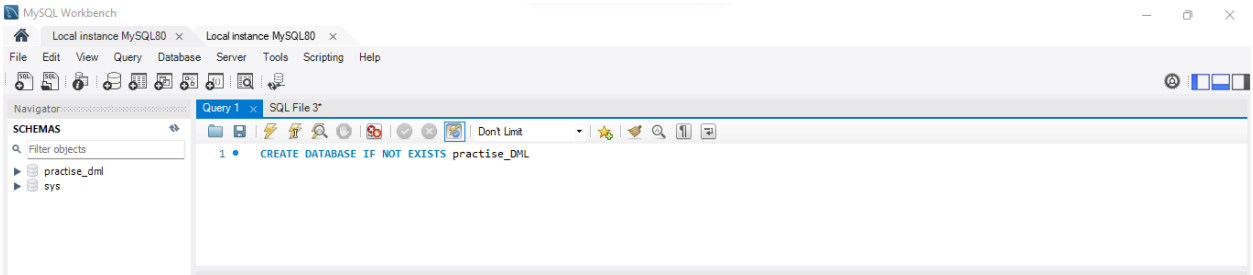
3. DDL in Detail

3.1 DDL Commands for Databases

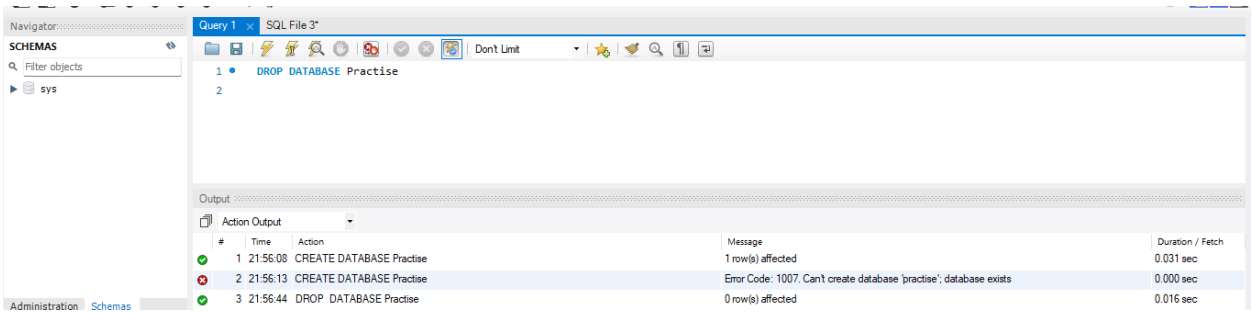
- `create` – to create a database



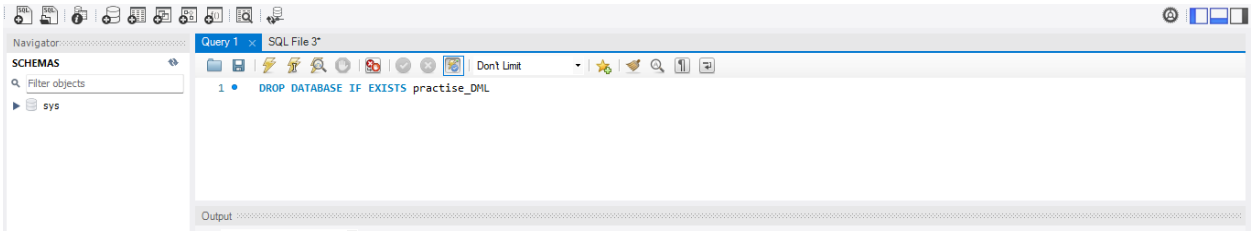
Better query:



- `drop` – to delete a database



Better query:

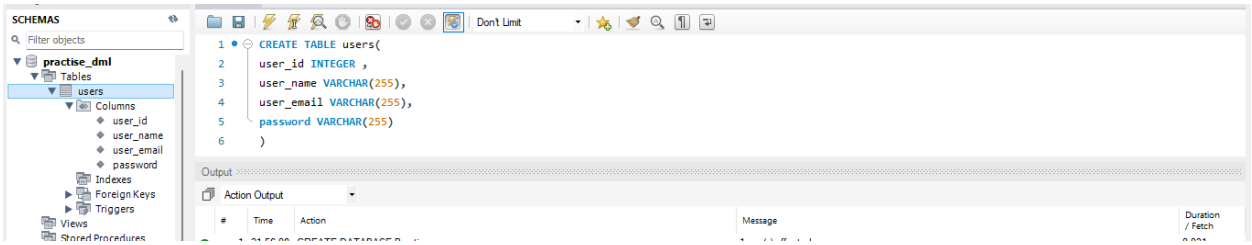


3.2 DDL Commands for Tables

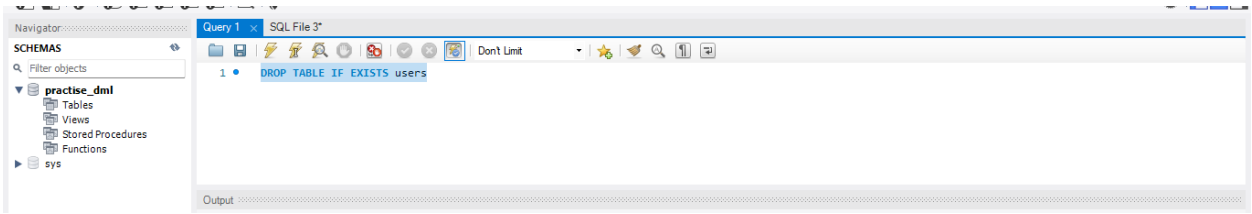
- Table name and attributes should be in **lowercase**
- `varchar` maximum range is **255 characters** (total 256)

Commands:

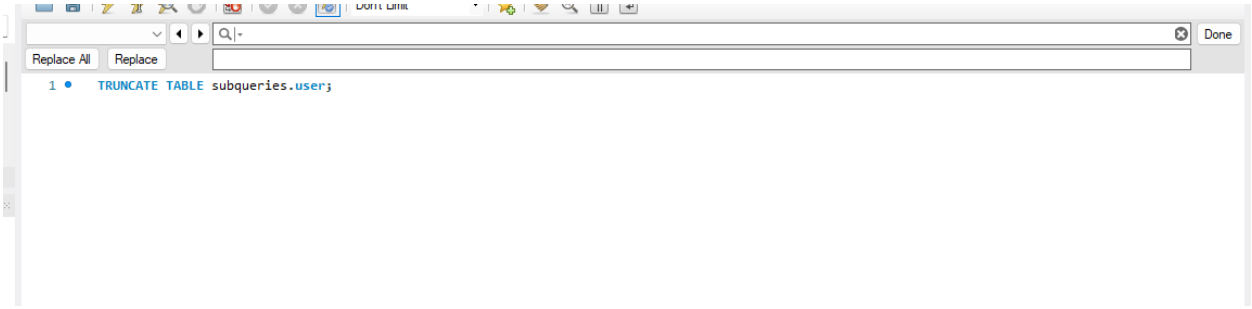
- `.create` – create table



- `.drop` – drop table along with rows



- `.truncate` – removes rows but keeps table structure



4. Data Integrity

Integrity of data ensures that data is **accurate, consistent, and complete**.
It measures the **reliability and truthfulness** of data and **prevents errors, corruption, or unauthorized changes**.

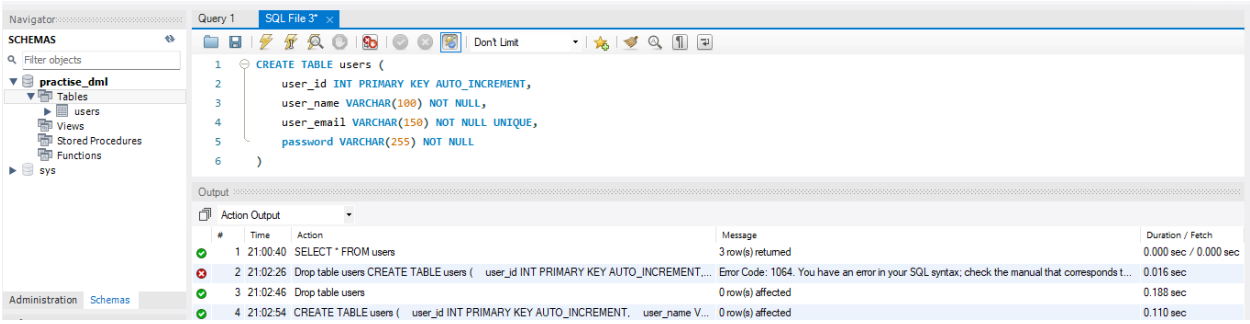
5. Methods to Maintain Data Integrity

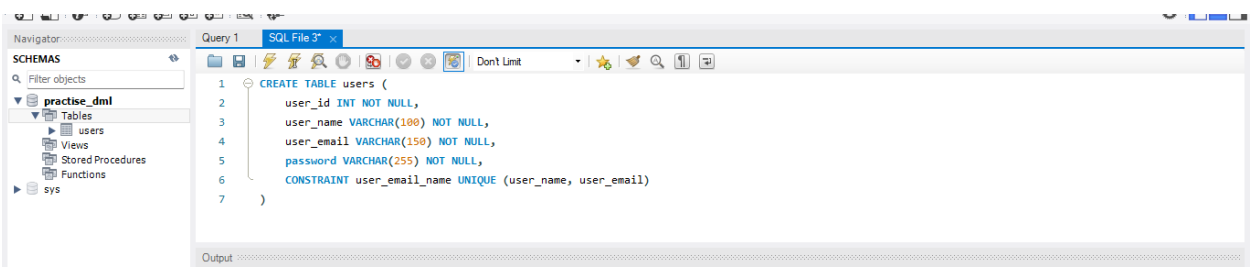
5.1. Constraints

Constraints are **rules applied to table columns** to maintain **accuracy, consistency, and reliability**.
They **prevent inconsistent or corrupted data**.

Common Constraints:

1. **NOT NULL** – e.g., roll number should not be null
2. **UNIQUE** – e.g., email should be unique
3. **PRIMARY KEY** – combines **NOT NULL** and **UNIQUE**, e.g., student ID
4. **AUTO INCREMENT** – e.g., automatically adds new user ID
5. **CHECK** – e.g., age > 18 for registration
6. **DEFAULT** – e.g., default country is Pakistan or current date
7. **FOREIGN KEY** – links tables





Referential Actions

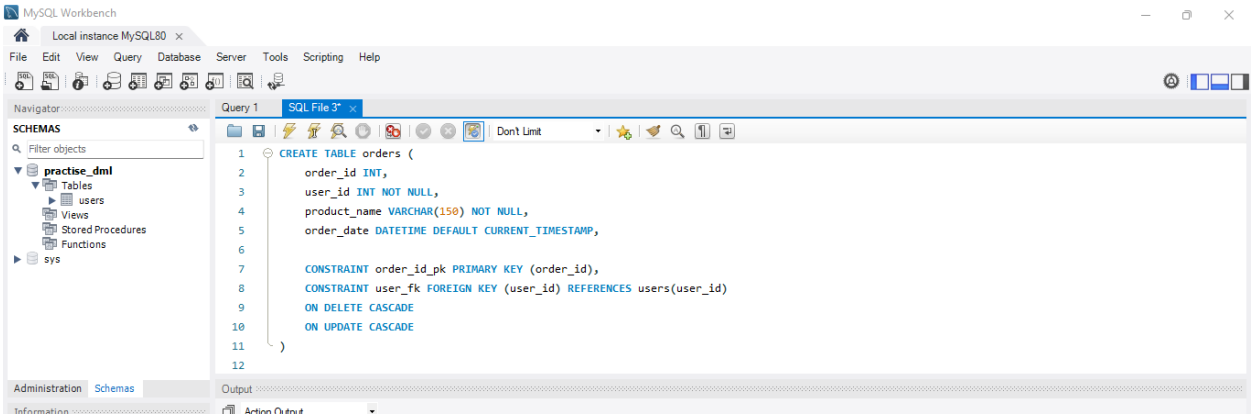
When tables are related, **update or delete in one table** can affect others.

Types:

1. **RESTRICT** – default; prevents deletion or update if dependent data exists
2. **CASCADE** – automatically updates/deletes dependent rows
3. **SET NULL** – deletes dependent row and sets NULL in foreign key column
4. **SET DEFAULT** – sets foreign key to default value, e.g., 0

Example:

- Two tables: `users` and `orders`
- `orders` has foreign key `user_id`
- Deleting `users` table without handling `orders` will throw error
- Referential actions protect **data integrity**



Benefits of Constraints

1. Apply the **same constraint** to multiple attributes at once
   - Example: Ensure **no two users have the same combination of name and email**
2. Remove constraints **without dropping the whole table**

5.2. Transactions

A **transaction** is a **sequence of database operations** treated as a **single unit of work**.
It ensures all operations are executed together, maintaining data integrity.

5.3. Normalization

Normalization is a design technique that:

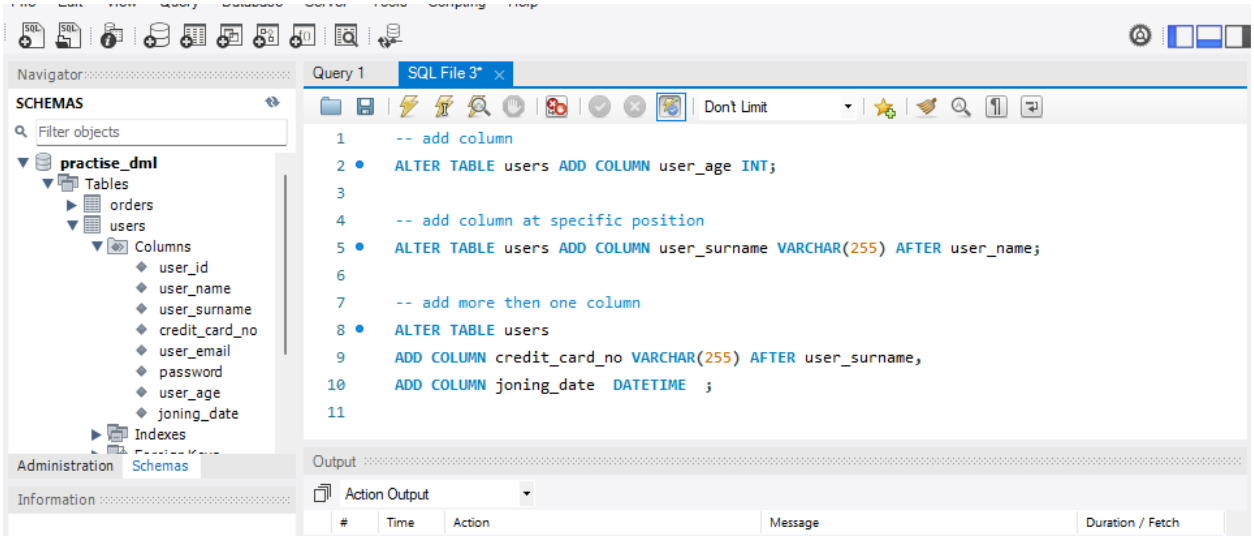
- Minimizes **data redundancy**
- Ensures **data consistency**
- Organizes data into **separate tables**

6. ALTER Command

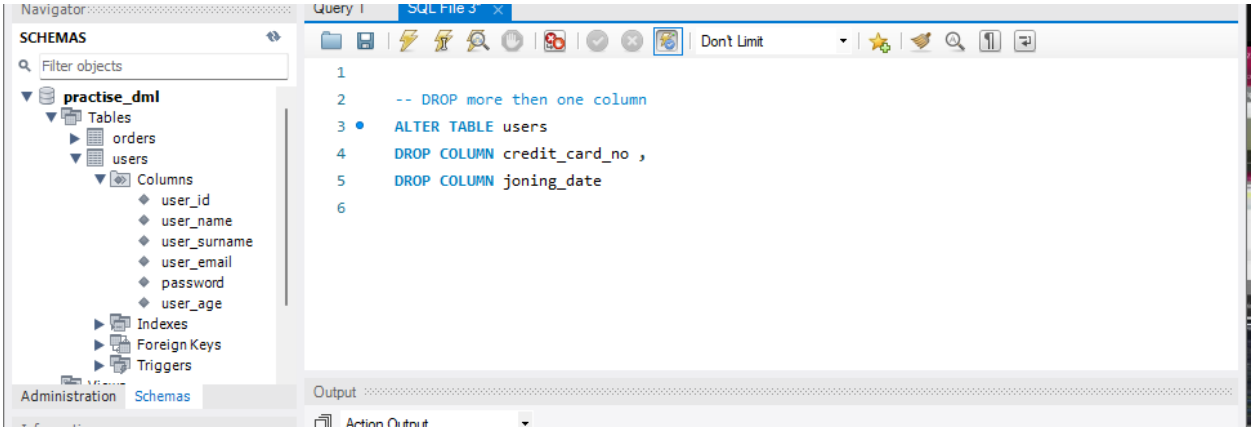
Used to **modify the structure of an existing table**.

Operations include:

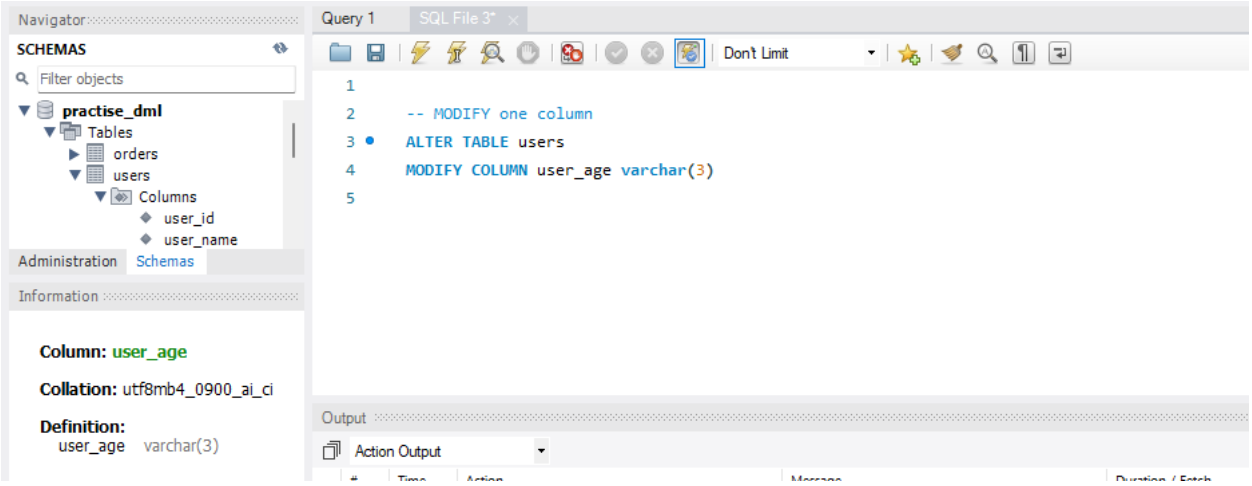
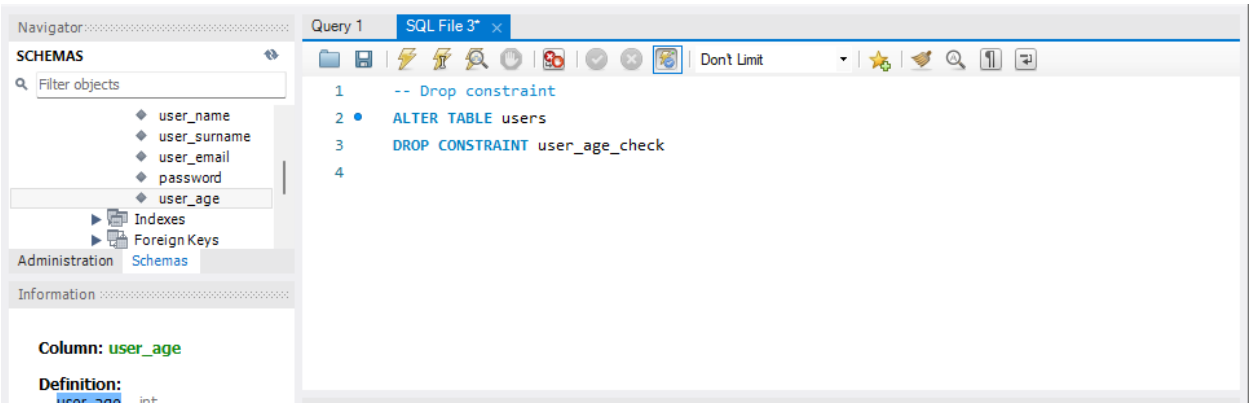
- **Add columns**



- **Drop columns**



- **Modify columns**



- **Add/Delete constraints**